

SDEV2501

Day 14 – Continuous Visualization

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Today’s objective

- Match chart types to data types (discrete vs. continuous) and analysis goals

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Agenda

- Review types of distributions
- Discuss properties of continuous distributions
- Showcase types of visualizations well-suited for continuous data
- Discuss how colour impacts our visualization

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REVIEW

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Review

- What is data viz? Why do we do it?

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CONTINUOUS DATA

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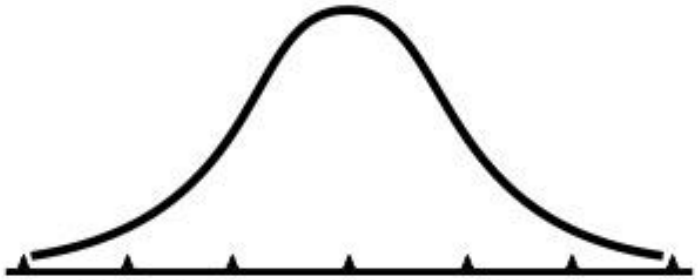
Continuous data vs continuous distributions

- **Continuous data** describes a variable that can take **any value within a range**.
  - e.g. temperature, height, elapsed time, distance, and weight.
- **Discrete data** takes **distinct countable** values.
  - e.g. number of students, number of orders, and number of errors.
- Real-world continuous measurements are often **rounded**.
  - e.g. Recording someone's height as 172 cm doesn't make height a discrete variable; it means the measurement was recorded to a particular precision.
- A **distribution** describes how observed values are spread across possible values.

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NORMAL DISTRIBUTIONS

- A type of continuous distribution, also known as a bell curve.
- Generally, mean = median = mode
- If data are normally distributed, we know:
  - ~68% of data are within 1SD
  - ~95% of data are within 2 SD
  - ~99.7% of data are within 3 SD
  - Generally, anything outside of 3SD is considered a **potential** outlier.
- Can **often** assume normal distribution for large datasets… but not always.
  - e.g. skewed, bimodal

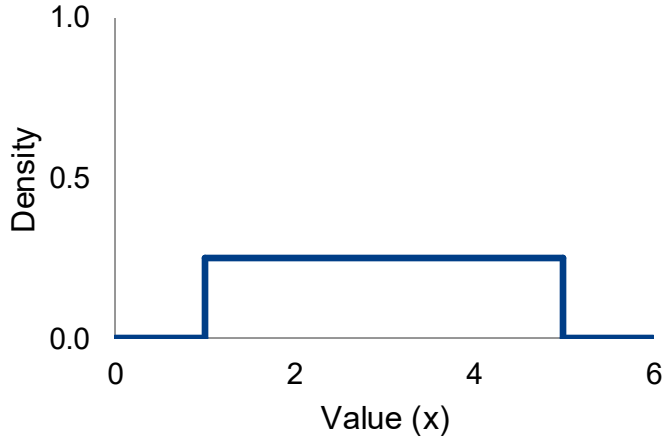




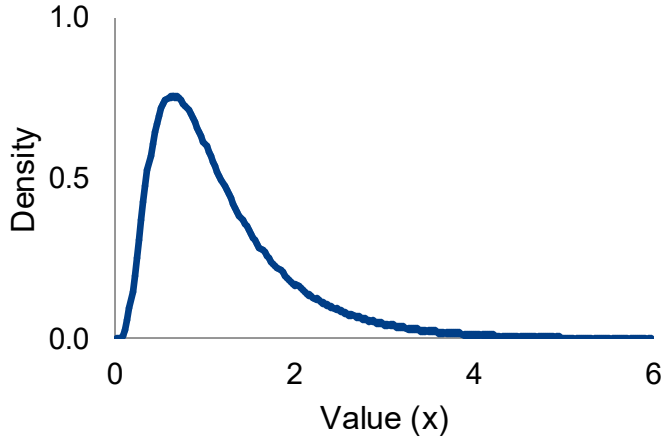
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Other Continuous Distributions

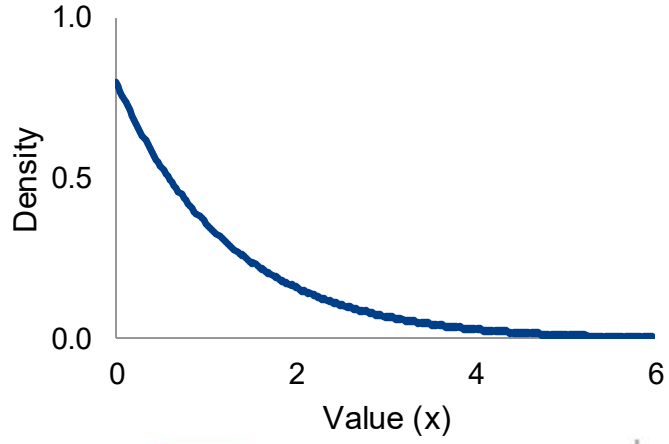
Uniform distributions
- Values spread evenly across a fixed range. Equal-sized intervals have equal probabilities.
- e.g. random number generator



Lognormal distributions
- An asymmetrical curve with a sharp peak & a long right-tail.
- e.g. distribution of human wealth



Exponential distributions
- A curve that starts at its max value and begins a smooth slide tapering to zero. Picture a playground slide.
- e.g. the amount of time a customer waits in line



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Real-Life Examples

- **Adult heights** within a reasonably **similar population** may be **approximately normal**.
- **Personal incomes** are often **right-skewed** because a relatively small number of people have very high incomes.
- **Delivery times** may be **right-skewed** because most deliveries arrive within a typical timeframe, but some experience substantial delays.
- **Exam scores** may be **bimodal** if two groups experienced different levels of preparation or the exam contained two difficulty levels.

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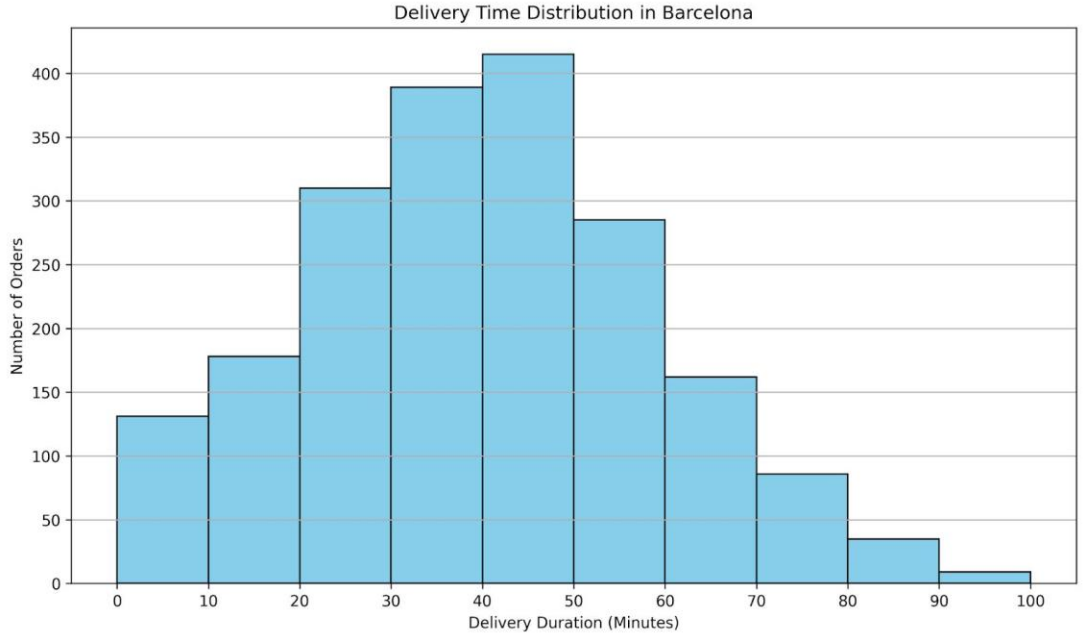
WHAT TYPES OF VIZ ARE BEST FOR CONTINUOUS DATA?

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HISTOGRAM

What does the distribution look like?

- Shows the **frequency** of numeric observations within intervals called **bins**.
- Useful for identifying **skewness**, clusters, gaps, and possible **outliers**.
- **Changing bin width** can change the apparent shape.

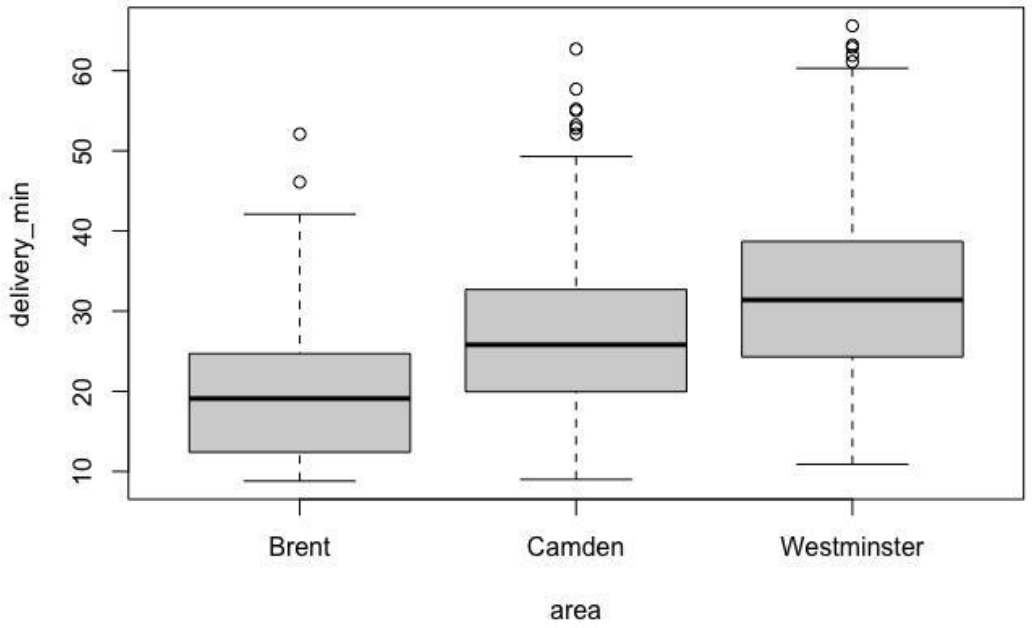


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BOX PLOT

How do groups **compare**?

- Summarizes the **median**, **quartiles**, and potential **outliers** according to the plotting convention.
- Useful for **comparing distributions** across categories.
- **Hides** much of the **detail** about clusters and the shape of the distribution.

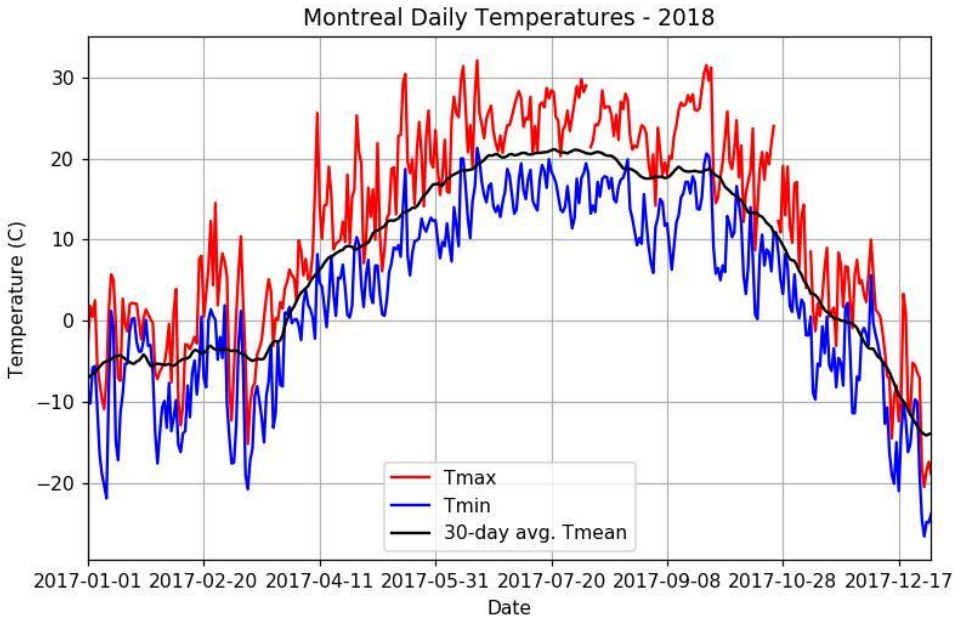


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LINE CHART

How does something **change** over **time**?

- Connects observations in a meaningful sequence, usually time.
- Makes **trends**, **cycles**, and **sudden changes** easier to see.
- Connecting unrelated categories can imply a continuity that does not exist.

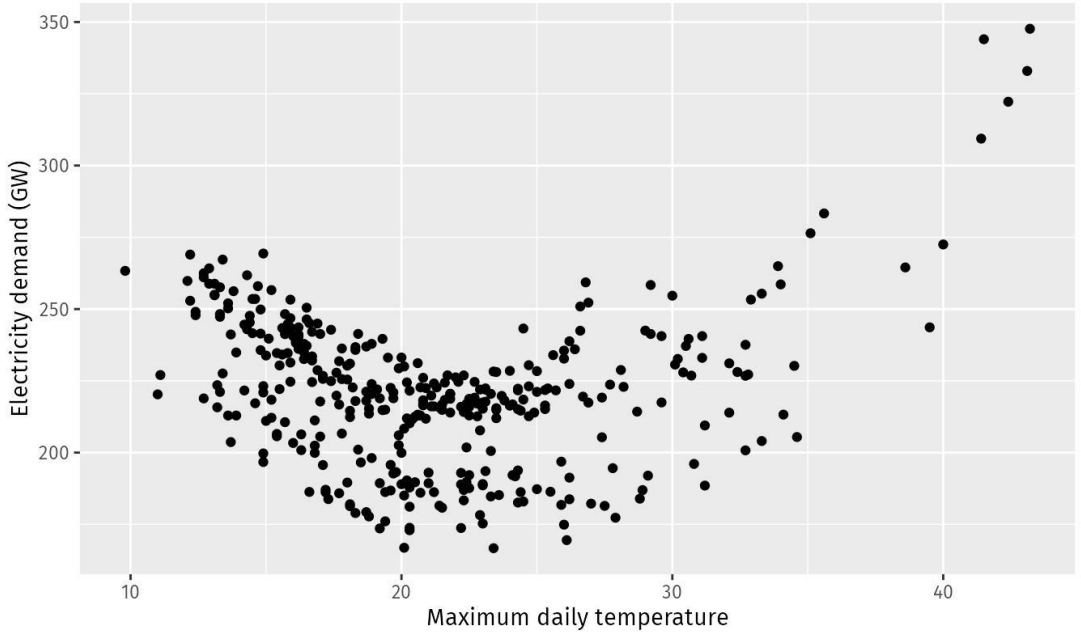


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SCATTER PLOT

Are two numeric variables **related**?

- Shows **paired** observations for two numeric variables.
- Helps reveal **relationships**, **clusters**, and unusual observations.
- A visible association does not establish **causation**.



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Other options

- **Density plot**: A smoothed estimate of a distribution's shape; its appearance depends on the smoothing bandwidth.
- **ECDF** (empirical cumulative distribution function): Shows the proportion of observations at or below each value. Useful for comparing distributions and answering percentile-style questions.
- **Violin plot**: Combines distribution shape with a compact group comparison; can be harder for beginners to interpret.
- **Error bars** or **interval plots**: Communicate uncertainty or variability around estimates, provided the intervals are clearly defined.

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Let’s look at some real-life examples.

Variable	Data type	Possible visualization
Daily temperature	Continuous	Line chart, histogram
Delivery time	Continuous	Histogram, box plot
# of deliveries	Discrete	Bar chart
Customer satisfaction (1-5)	Ordinal	Ordered bar chart
Monthly revenue	Numeric (often assumed continuous)	Line chart, histogram

A **continuous** variable does not have to follow a normal distribution.

e.g. Temperature could be approximately normal in one context and strongly skewed in another.

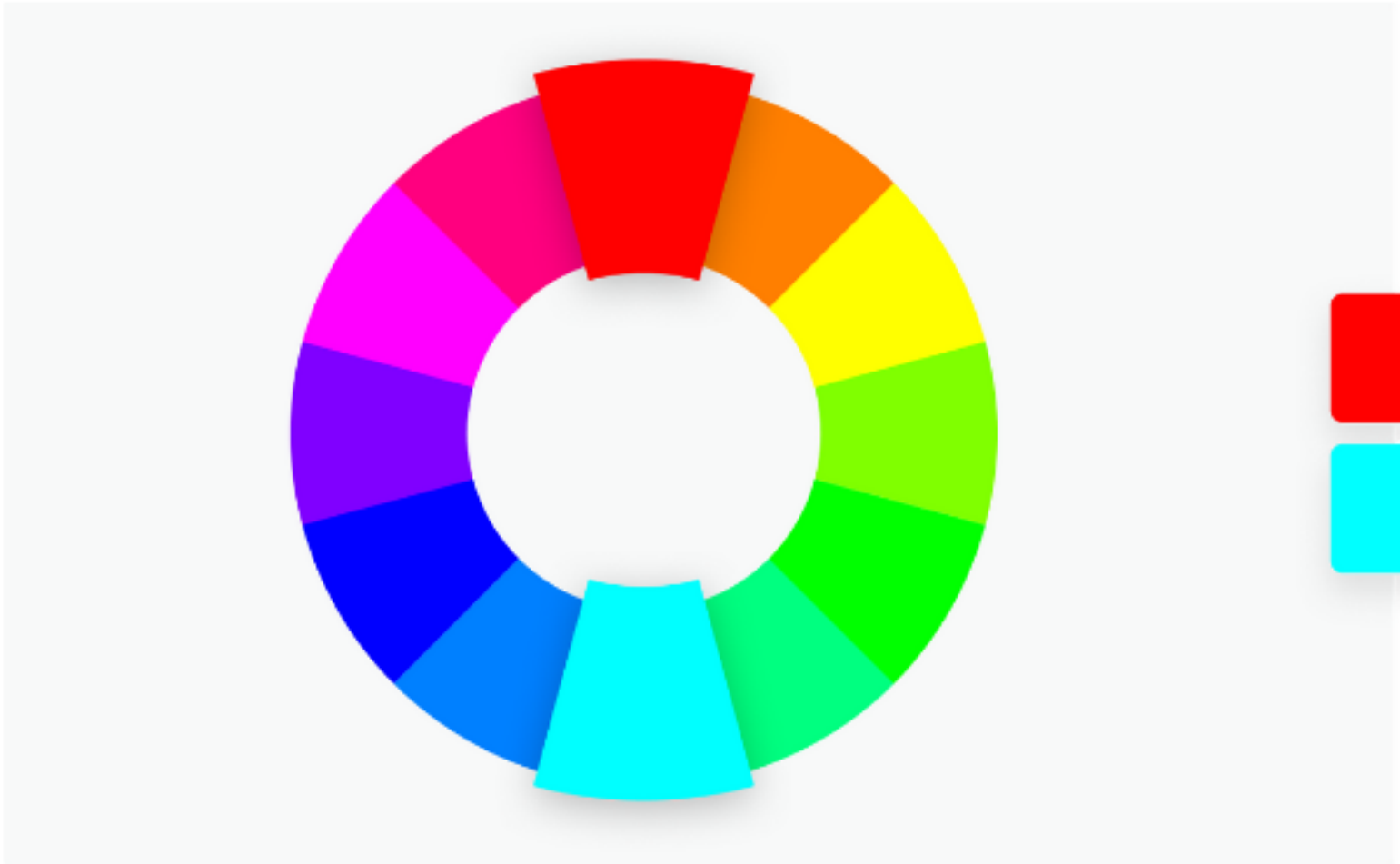
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COLOUR THEORY

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Complementary

Two colors that are on opposite sides of the color wheel. This combination provides a high contrast and high impact color combination – together, these colors will appear brighter and more prominent.



Monochromatic

Three shades, tones and tints of one base color. Provides a subtle and conservative color combination. This is a versatile color combination that is easy to apply to design projects for a harmonious look.



Analogous

Three colors that are side by side on the color wheel. This color combination is versatile, but can be overwhelming. To balance an analogous color scheme, choose one dominant color, and use the others as accents.



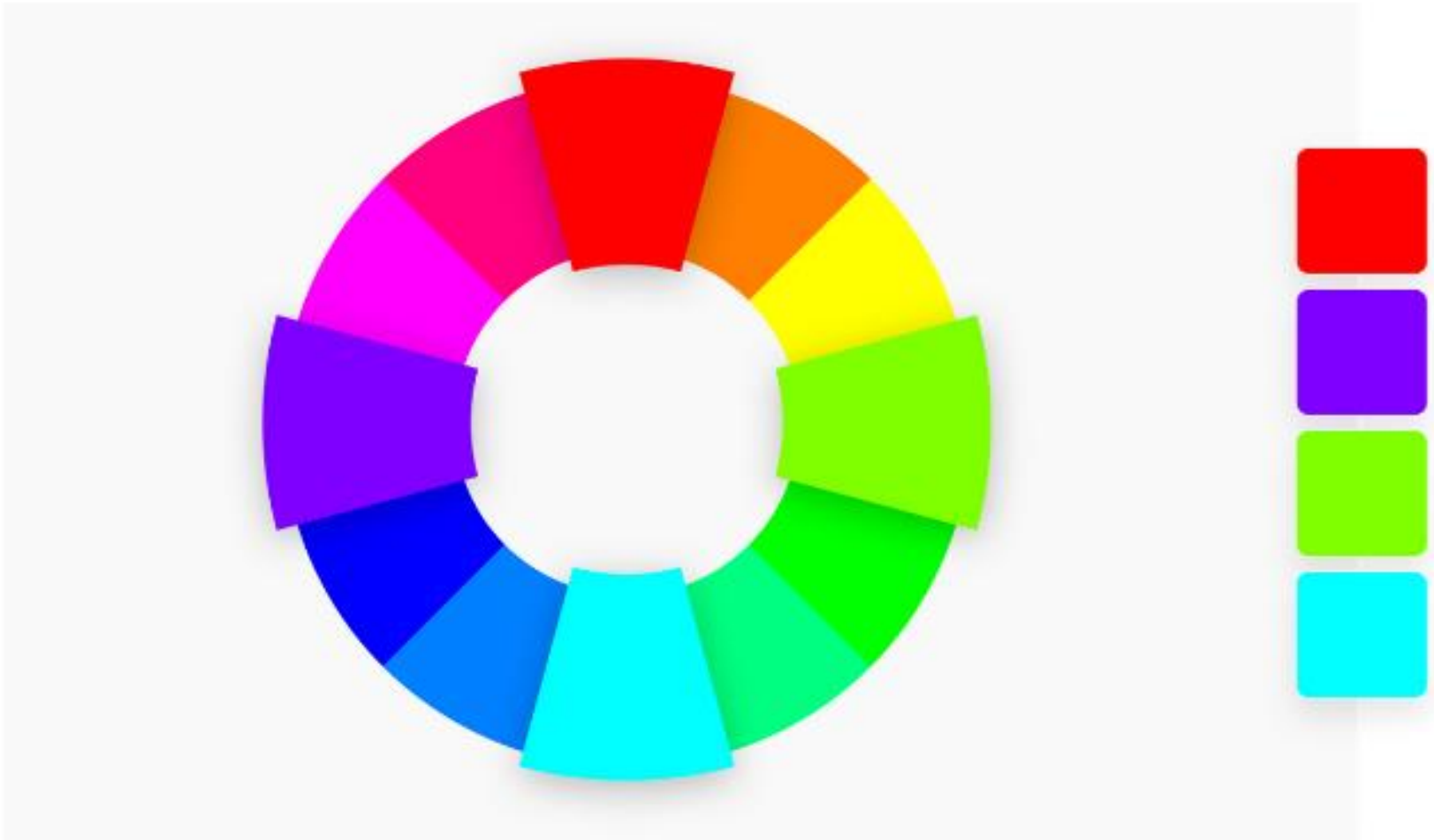
Triadic

Three colors that are evenly spaced on the color wheel. This provides a high contrast color scheme, but less so than the complementary color combination — making it more versatile. This combination creates bold, vibrant color palettes.



Tetradic

Four colors that are evenly spaced on the color wheel. Tetradic color schemes are bold and work best if you let one color be dominant, and use the others as accents. The more colors you have in your palette, the more difficult it is to balance,



Color Schemes with #1e90ff



Complementary Color



Analogous Color



Split Complementary Color



Triadic Color



Tetradic Color



Monochromatic Color

How to choose a palette

- **Sequential palettes**
  - Use a **progression** from light to dark, or low intensity to high intensity, when values have a **natural order** from low to high.
  - e.g. population density by neighbourhood.
- **Diverging palettes**
  - Use two **contrasting** colour directions around a **meaningful** midpoint.
  - e.g. temperature differences from the seasonal average, where below-average and above-average values are important.
- **Qualitative palettes**
  - Use **distinct** colours to distinguish **categories** without implying an order or magnitude.
  - e.g. comparing product types.

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The job of colour

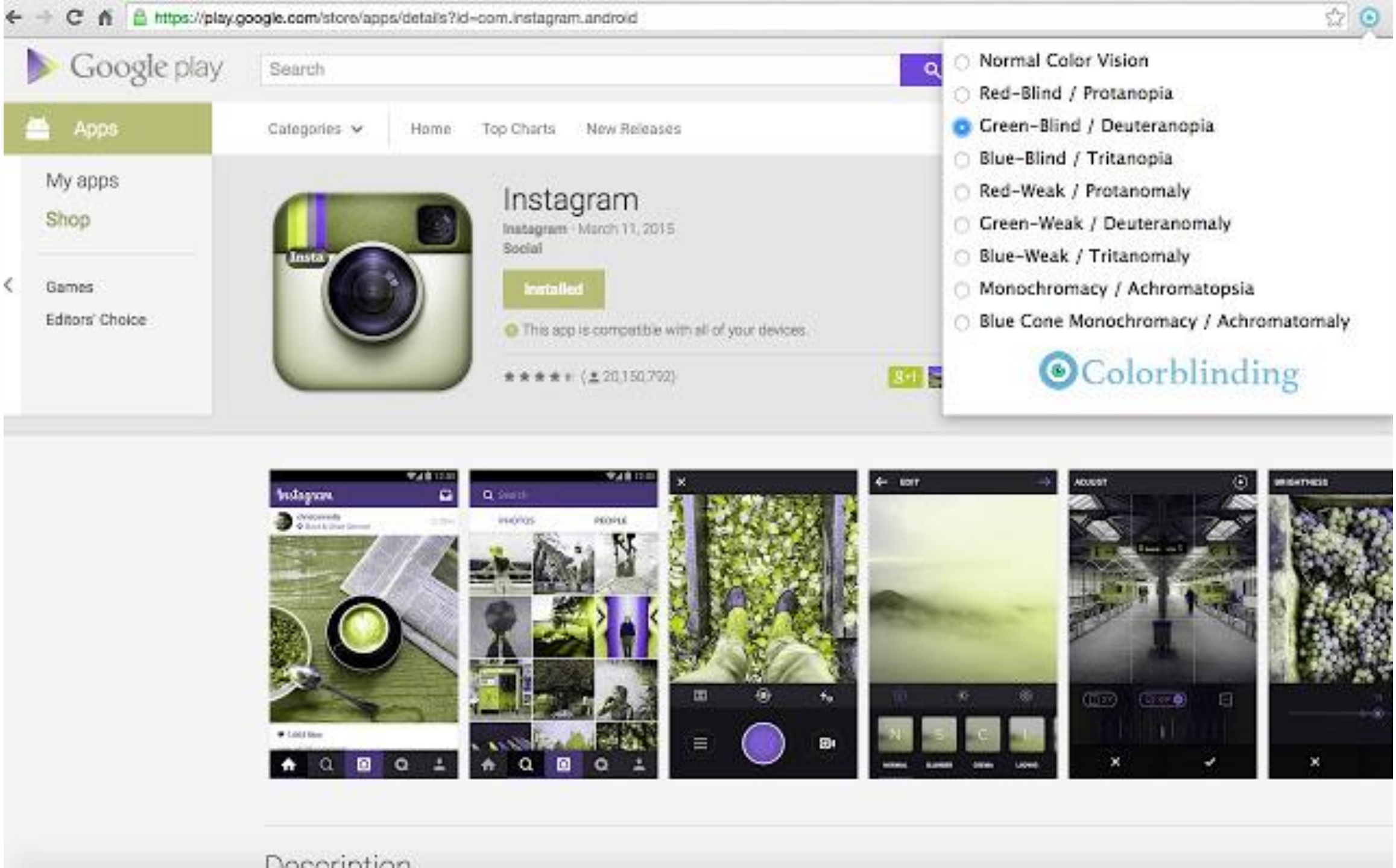
- Use **colour** to encode a **variable**, **highlight** a finding, group related information, or establish **hierarchy**.
- Avoid assigning a different colour to every mark when the colours have no analytical meaning.
- Use a **neutral** base palette and reserve **stronger colour** for the **most important** information.
- **Keep colour meanings consistent** across related charts. If blue means "below target" on one chart, don't casually use it to mean "above target" on the next.
- Don't use red and green as the only distinction between two important values.

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Accessibility

- Never make **colour** the only way to communicate a **distinction** that matters.
- **Pair colour** with labels, symbols, line styles, shapes, or patterns where appropriate.
- Check **contrast** between text and its background, not just whether the colours look different.
- Test charts in grayscale and with colour-vision simulations.
- Avoid relying exclusively on **red versus green** to communicate success versus failure.
- **Consider the entire chart**: small markers, thin lines, overlapping series, and low-contrast gridlines can all reduce readability.

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Google play
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Editors' Choice
Categories
Home
Top Charts
New Releases
Instagram
Instagram · March 11, 2015
Social
Installed
This app is compatible with all of your devices.
Normal Color Vision
Red-Blind / Protanopia
Green-Blind / Deuteranopia
Blue-Blind / Tritanopia
Red-Weak / Protanomaly
Green-Weak / Deuteranomaly
Blue-Weak / Tritanomaly
Monochromacy / Achromatopsia
Blue Cone Monochromacy / Achromatomaly
Colorblinding

What colour is this dress?



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> Who in the rainbow can draw the line where the violet tint ends and the orange tint begins? Distinctly we see the difference of the colors, but where exactly does the one first blendingly enter into the other? So with sanity and insanity.
>
> —Herman Melville, *Billy Budd*

> Orange, red? I don't know what to believe anymore!
>
> —Anonymous, Color Survey

shark invested water blue

white if it had a wine spill on itself and let it dry for a few days and then tried to wash it but it just left it that awful color that foreverr remains a reminder that grandma needs to be cut off after the third glass

we are a collaborative icecream cone!!!!!

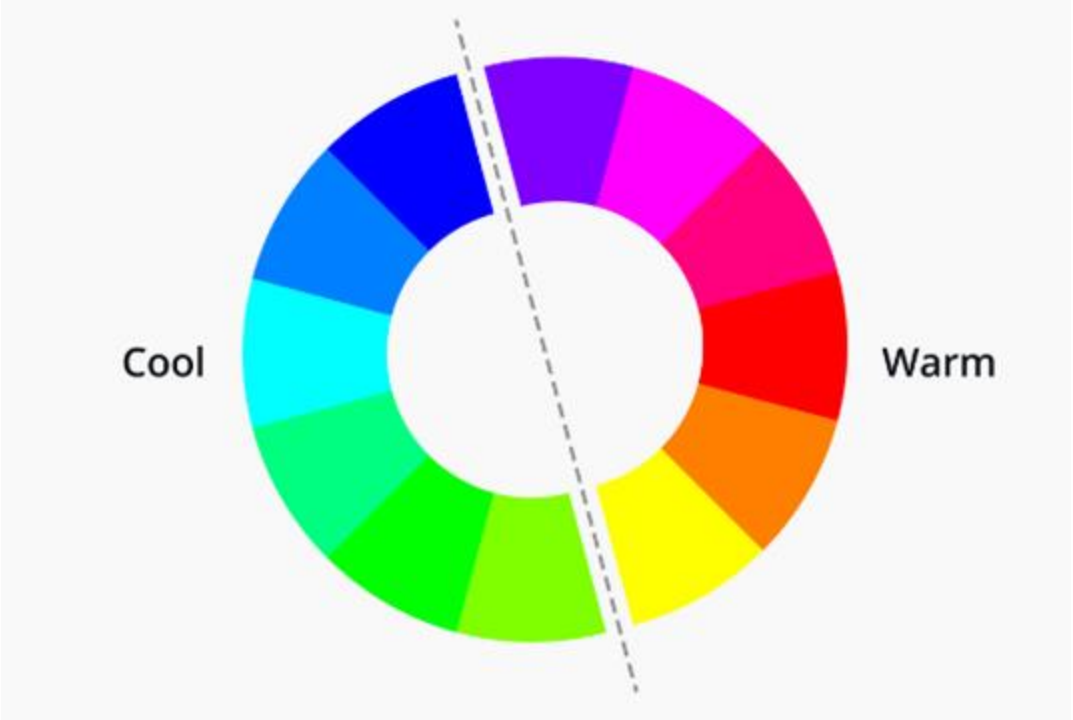
day 3: sanity lost, colors keep changing but they keep staying the same...keep seeing this green, this slightly different green, mocking me...studying me...this ms green...what do you want mr green

really? this color again? i have nothing against colors personally, but this one just stands out from the rest as unusually unnattractive. i almost feel sad for it, but it made the decision to be that color so it has to find a way to deal with it

Colours & Emotions

Warm colours bring to mind coziness, energy, and warmth.

Cool colours are associated with serenity, isolation, and coolness.



Group Activity: Colours & Emotions

- What **feelings** are associated with this colour?
- Does it have any physical **effects**?
- Does it have different **cultural meanings**?
- What is this colour **best used for**?
- Find (or create) an **example**.

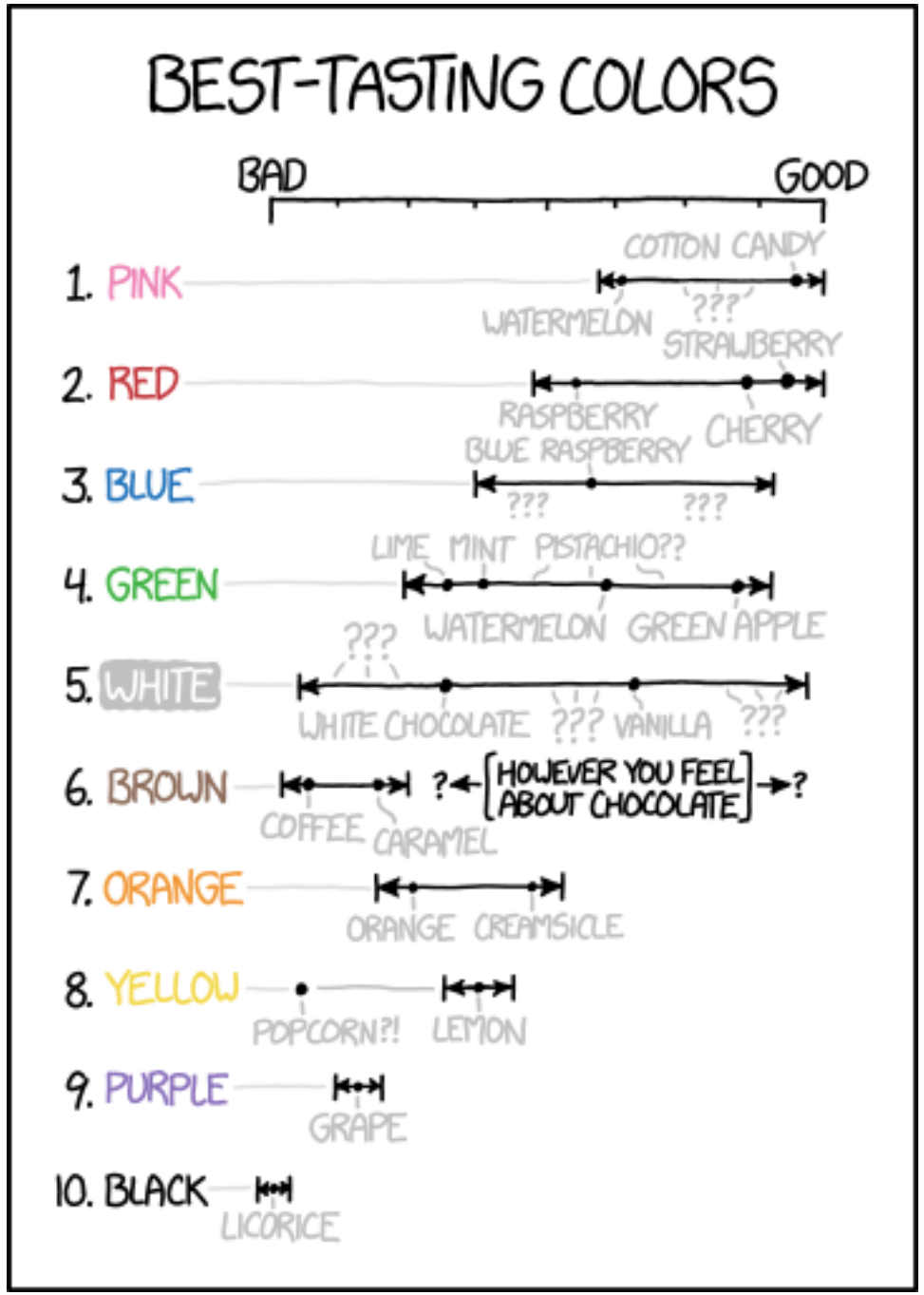
1. Red
2. Orange
3. Yellow
4. Green
5. Blue
6. Purple
7. Pink
8. Black
9. White
10. Gray
11. Brown

Colour Resources

- The meaning of colour
  - color-theory-for-designers-part-1-the-meaning-of-color
  - canva.com/colors/color-meanings
- Palette generation:
  - colorhexa.com
  - carto-colors
  - coolors.co
  - colorgorical
  - colorbrewer2
  - iwanthue
  - colorhunt
- The colour of anything

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BEST-TASTING COLORS



Guided demo

- Load CSV, brief inspection
- Create a basic histogram with plt.hist()
  - Set bins, add title & axis labels
- Improve: bin width, figsize, set colour, add units
- Create a boxplot with plt.boxplot()
  - Compare, explain the median/quartiles/whiskers
- Create a scatter plot with plt.scatter() w/ appropriate labels
- Identify limitations of each viz

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Individual Practice

- Given one dataset, create 2 different types of charts.
  - Each should answer a different question e.g. distribution vs relationship

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Closing

- Find 3 real-life examples of distributions using continuous data.
  - What story do they tell?
  - How are values represented? e.g. colour, position, area…
  - What is the most important pattern?
  - How could you improve the viz?

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What’s next?

- Visualization of discrete data

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